



Shawanwit Poomsa-ad

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Date of birth: 15 August 2004 **Nationality:** Thai

Languages: Thai (native), English (fluent)

Education

- Aug. 2024 – Pres. **Bachelor of Science in Mathematics , Minor in Physics**
Purdue University, West Lafayette, IN (USA)
GPA: 3.70/4.00
Relevant coursework: Representation Theory, Honours Galois Theory; Data Science Laboratory in Probability; Abstract Linear Algebra; Honours Real Analysis I and II; Honours Abstract Algebra; Fusion Categories in Physics; Astroparticle Physics;
- Jun. – Aug. 2025 **Study Abroad Semester — Budapest Semesters in Mathematics**
Hosted by Eötvös Loránd University and the Alfréd Rényi Institute of Mathematics, Budapest, Hungary
Coursework: Number Theory; Complex Functions.

Research Experience

- Sep. 2025 – Pres. **Undergraduate Researcher — Machine Learning and Knot Theory**
Experimental Mathematics Laboratory, Purdue University
Supervisor: Prof. Thi Hanh Vo.
 - Developing a Kolmogorov-Arnold Network (KAN) to classify mathematical knots by bridge number using large-scale knot datasets (>1M labelled examples).
 - Exploring ways to use similar methods for other knot invariants.
- Jan. – May 2025 **Undergraduate Researcher — Time-Domain Astrophysics**
The Physics Datamine, Purdue University
Supervisor: Prof. Danny Milisavljevic.
 - Led a 5-person team to build a Python filtering algorithm detecting rebrightening patterns in astrophysical data.
 - Preprocessed and optimised a 10,000-entry forced photometry dataset of noisy supernovae data using wavelet analysis.
 - Project considered for continuation in mathematical modelling of noisy astronomical signals, in anticipation of the Vera C. Rubin Telescope.
 - Selected to present poster and results at the Purdue Research Expo.
- Aug. – Dec. 2024 **Undergraduate Researcher — High-Energy Particle Physics**
The Physics Datamine, Purdue University
Supervisor: Prof. Andreas Jung.
 - Developed a signal selection algorithm in Python to analyse the CMS open dataset and reconstruct the Higgs boson signal from event data.
 - Selected to present poster and results at the Purdue Research Expo.

Skills

Computing Python, C, C++, Julia, R, MATLAB, Lean, L^AT_EX